

Module 3: Data and Information Management for the Web

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Objectives/Learning Outcomes

- Evaluate web data architectures
- Design enterprise-level data models
- Assess API-based data integration
- Analyze governance and compliance risks
- Recommend appropriate storage technologies

Data → Information → Knowledge → Wisdom

- Data = raw facts
- Information = processed data
- Knowledge = contextualized information
- Wisdom = strategic decision

Data as Strategic Asset

- Amazon – personalization
- Netflix – recommendation systems

Is Amazon a retailer or a data company?

Enterprise Data Architecture Layers

- Presentation
- Application
- Data
- Infrastructure

What happens if business logic is placed in the database?”

Centralized vs Distributed Data

Centralized

Easier control

Single failure point

Distributed

Scalable

Complex consistency

To maintain complex consistency without a single "brain," systems use specific protocols:

- **Two-Phase Commit (2PC):** A "coordinator" asks all nodes if they are ready to commit. If everyone says yes, the coordinator sends a "do it" signal.
- **Sagas:** Instead of one big transaction, you use a sequence of local transactions. If one fails, the system runs "compensating transactions" (undoing previous steps) to restore balance.
- **Quorum-based Voting:** To ensure a complex write is "official," a majority (quorum) of nodes must agree on the new state before it's considered consistent.

Three-Tier Architecture

- Client → App Server → Database

Microservices & Containers

- Docker - is a platform that packages an application and all its dependencies into a standardized, lightweight **container** to ensure it runs consistently across any computing environment.
- Kubernetes - is a platform that automates the deployment, scaling, and management of containerized applications across a cluster of servers.

ADVANCED DATA MODELING

Conceptual, Logical, Physical Models

- Conceptual = business view
- Logical = normalized schema
- Physical = indexes, storage engines

ER Modeling

- Entities, Attributes, Relationships

Relational vs NoSQL

- PostgreSQL
- MongoDB

Which one is more scalable?

Indexing & Query Optimization

Can too many indexes reduce performance?

WEB DATA INTEGRATION & APIs

- Modern web systems rely on APIs.

API-Driven Architecture

- **REST** (Representational State Transfer) is to provide a **standardized, stateless way** for different computers to talk to each other over the internet using the existing rules of the web (HTTP).
- **GraphQL** is a query language and server-side runtime specification that allows clients to request exactly the data they need from a single endpoint, mirroring the structure of the returned JSON response.

The best way to compare **REST** and **GraphQL** is to think about how you order food. **REST** is like a fixed menu at a restaurant—you get exactly what's on the plate. **GraphQL** is like a buffet—you take exactly what you want, in the portions you need, and nothing more.

JSON & Data Exchange

- JSON as standard data exchange format.

Enterprise Integration Patterns

- ETL (Extract-Transform-Load)
- Data Warehousing
- Event-driven architecture
- Message queues (e.g. RabbitMQ)

Apache Kafka is an open-source, distributed event streaming platform designed to handle real-time data feeds with high throughput and low latency. Originally created at LinkedIn and now maintained by the Apache Software Foundation, it enables applications to publish, subscribe to, store, and process streams of records at scale.

DATA GOVERNANCE, SECURITY & COMPLIANCE

Data Governance Framework

Components:

- Data ownership
- Data stewardship
- Data quality management
- Metadata management

Component

Real-World Application

Data Ownership

The Chief Medical Officer (CMO) owns the patient data. They don't touch the database, but they sign off on the privacy policies and decide that "Only doctors and nurses assigned to a patient may view their full history."

Data Stewardship

The Lead Health Informatics Officer acts as the Steward. When a doctor complains that "Birth Date" is being entered in three different formats across departments, the Steward creates a rule: "All dates must follow ISO 8601 (YYYY-MM-DD)."

Data Quality Management

An Automated Validation Script runs every time a nurse hits 'Save'. If a blood pressure reading is entered as "2000/100" (an obvious typo), the system blocks the entry and flags it for review to ensure **Accuracy** and **Validity**.

Metadata Management

The Hospital Data Catalog provides the context. A researcher looking at a "Cholesterol" column can check the metadata to see if the units are in \$mg/dL\$ or \$mmol/L\$, and which specific lab equipment generated that number.

Web Data Security Risks

- SQL Injection
- XSS (Cross-Site Scripting)
- CSRF (Cross-Site Request Forgery)
- API abuse

Attack	Target	Language Used	Analogy
SQLi	Database	SQL	Tricking the File Clerk
XSS	Other Users	JavaScript	Putting poison in the water cooler
CSRF	User's Identity	HTTP Requests	Forging someone's signature
API Abuse	The Server	Anything	Knocking on the door 1 million times

Security Controls

- Encryption at rest
- Encryption in transit (TLS)
- Role-Based Access Control (RBAC)
- Audit logging

Cloud Databases

- Amazon RDS
- Google Cloud Firestore

Benefits:

- Scalability
- Elastic storage
- Managed services

Big Data & Web Systems

Concepts:

- Data lakes
- Real-time analytics
- Stream processing

Apache Hadoop?

Case Analysis

A Philippine university plans to:

- Move student records to cloud
- Provide mobile API access
- Integrate LMS & SIS
- Comply with Data Privacy Act

Questions:

- What architecture would you propose?
- What governance framework is needed?
- What database model is most suitable?
- What risks must be mitigated?

Key Takeaways

- Data is strategic asset
- Architecture decisions impact scalability
- Governance is mandatory
- Security is non-negotiable
- Web systems are increasingly API-driven

Thank You.